HUJI · 67731 · OPTIMIZATION · 2026-27

Week 01

Vectors, Matrices, and Linear Systems

norms · Cauchy-Schwarz · eigenvalues · positive definiteness · Cholesky · conditioning · LU

CONTENTS

PART A · LECTURE 1

1 Vectors and norms 2
2 Matrices 4
3 Positive definiteness 6
4 Matrix norms 8
5 Linear systems 9
6 Solving with Cholesky 10

PART B · RECITATION 1

7 Exercise 1.1: three norms of three vectors 12
8 Exercise 1.2: ℓ_2 is a norm 12
9 Exercise 1.3: sharp equivalence constants 13
10 Exercise 1.4: the condition number 14
11 Exercise 1.5: the matrix inner product 15
12 Exercise 1.6: entries of a PSD matrix 16
13 LU in the notes and Exercise 1.7 17
14 Exercise 1.8: the PSD order (Moed C 2024) 17

PART C · RECITATION 1 HANDOUT

15 What do we want? 19
16 Why are triangular systems easy? 19
17 The idea: LU is exactly Gaussian elimination 19
18 The algorithm (without row swaps) 20
19 Example 1: factoring a 3×3 matrix 20
20 Solving $Ax = b$ with the factorization 21
21 Why do we need row swaps (P)? 21
22 What is the matrix P? 22
23 The algorithm with swaps 23
24 Example 2: factoring with a row swap 23
25 Solving $Ax = b$ with P 24
26 Summary of LU 25

WEEK 01

27 Practice problems with full solutions 26
28 Cheat sheet 27

PART A · LECTURE 1

Vectors, Matrices, and Linear Systems

SECTION 01

Vectors and norms

Throughout the course we work with vectors $x \in \mathbb{R}^n$, i.e. lists of n real numbers $x = (x_1, \dots, x_n)$. To optimize we must be able to say how *big* a vector is (how far a guess is from the answer, how large an error is). That is the job of a norm.

DEF · NORM

A norm $\|\cdot\|$ measures the "magnitude" of a vector, i.e. its distance from the origin. It must satisfy three axioms:

1. Positive definiteness: $\|x\| \geq 0$, with equality *if and only if* $x = 0$.
2. Absolute homogeneity: $\|\alpha x\| = |\alpha| \, \|x\|$ for every scalar α.
3. Triangle inequality: $\|a + b\| \leq \|a\| + \|b\|$.

WHAT THE AXIOMS SAY IN WORDS

(1) Only the zero vector has length 0. (2) Doubling a vector doubles its length, and flipping its sign changes nothing. (3) Going from 0 to $a + b$ directly is never longer than going via a: a detour cannot be a shortcut.

> 1.1 The three common norms

ℓ_1, ℓ_2, ℓ_∞

$$\|x\|_2 = \sqrt{\textstyle\sum_i x_i^2} \qquad (\ell_2, \text{ measures energy})$$
$$\|x\|_1 = \textstyle\sum_i |x_i| \qquad (\ell_1, \text{ good for sparsity})$$
$$\|x\|_\infty = \max_i |x_i| \qquad (\ell_\infty, \text{ worst case})$$

EXAMPLE · ONE VECTOR, THREE LENGTHS

$x = (3, -4)$: $\|x\|_2 = \sqrt{9 + 16} = 5$ (the straight-line distance), $\|x\|_1 = 3 + 4 = 7$ (walking along a city grid), $\|x\|_\infty = \max(3, 4) = 4$ (the single worst coordinate).

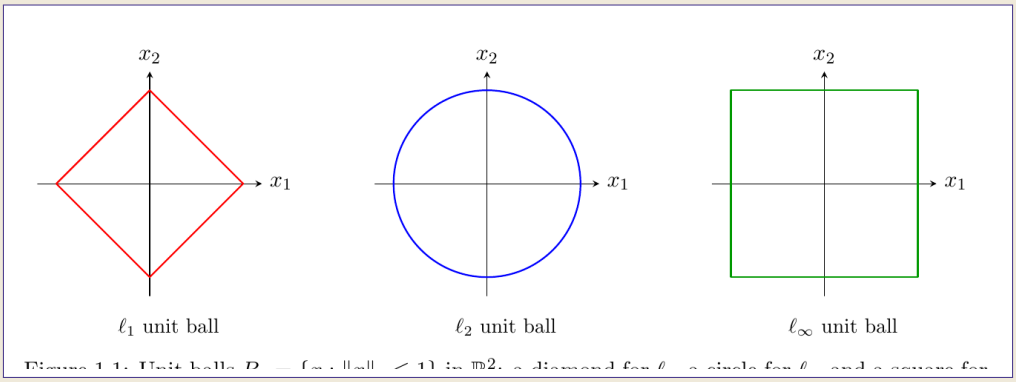


Figure 1.1 of the notes: the unit balls $B_p = \{x : \|x\|_p \le 1\}$ in $\mathbb{R}^2$, a diamond for ℓ_1, a circle for ℓ_2, a square for ℓ_∞. They are nested, $B_1 \subseteq B_2 \subseteq B_\infty$, which is the picture behind $\|x\|_\infty \le \|x\|_2 \le \|x\|_1$.

DEF · THE "ℓ_0 NORM"

$\|x\|_0 =$ the number of nonzero entries of x. Example: $\|(3,0,0,-1)\|_0 = 2$. It is the natural measure of sparsity and returns in Chapter 13.

PITFALL · ℓ_0 IS NOT A NORM

It fails homogeneity: $\|2x\|_0 = \|x\|_0$, not $2\|x\|_0$. For $x = (1,0)$: $\|2x\|_0 = 1 \neq 2 = 2\|x\|_0$.

> 1.2 All norms are equivalent

EQUIVALENCE OF NORMS

On $\mathbb{R}^n$ every two norms are related by constant factors that do not depend on x. For our three norms (proved in Exercise 1.3):

$$\|x\|_\infty \le \|x\|_2 \le \sqrt{n}\,\|x\|_\infty, \qquad \|x\|_2 \le \|x\|_1 \le \sqrt{n}\,\|x\|_2.$$

WHY THIS MATTERS

If an algorithm makes the error small in one norm, it is small in every norm. The constants can depend on the dimension n, though, so in high dimension the choice of norm does matter quantitatively.

> 1.3 The Cauchy-Schwarz inequality

THEOREM 1.2: CAUCHY-SCHWARZ

For all $a, b \in \mathbb{R}^n$: $|a^T b| \le \|a\|_2\,\|b\|_2$, with equality if and only if a and b are linearly dependent.

Proof, step by step (the lecture's proof uses an optimization problem):

1. If $b = 0$ both sides are 0. Assume $b \neq 0$.
2. For any scalar c, a norm squared is never negative: $0 \le \|a - cb\|_2^2$. So also $0 \le \min_c \|a - cb\|_2^2$.

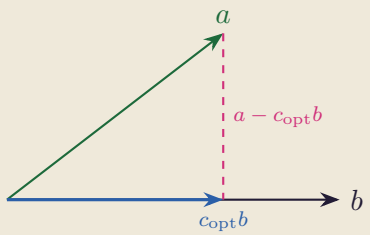
3. Expand: $\|a - cb\|_2^2 = (a - cb)^T(a - cb) = \|a\|^2 - 2c\,a^Tb + c^2\|b\|^2$. This is a parabola in c.

4. Minimize: set the derivative $-2a^Tb + 2c\|b\|^2$ to 0, giving $c_{\text{opt}} = \dfrac{a^Tb}{b^Tb}$.

5. Plug in: $\|a\|^2 - 2\dfrac{(a^Tb)^2}{\|b\|^2} + \dfrac{(a^Tb)^2}{\|b\|^2} = \|a\|^2 - \dfrac{(a^Tb)^2}{\|b\|^2}$. So

$$0 \le \|a\|_2^2 - \frac{(a^Tb)^2}{\|b\|_2^2} \quad \Longrightarrow \quad (a^Tb)^2 \le \|a\|_2^2\|b\|_2^2.$$

6. Take square roots. Equality holds exactly when $\min_c\|a - cb\| = 0$, i.e. $a = cb$ for some c: a and b are linearly dependent.



$c_{\text{opt}}b$ is the projection of a on the line of b. The leftover $a - c_{\text{opt}}b$ has length ≥ 0; that is the whole proof.

EXAMPLE · CHECKING IT

$a = (1, 2)$, $b = (3, 1)$: $|a^Tb| = 3 + 2 = 5$, and $\|a\|\|b\| = \sqrt{5}\sqrt{10} = \sqrt{50} \approx 7.07$. Indeed $5 \le 7.07$. With $a = (2, 4) = 2 \cdot (1, 2)$ and $b = (1, 2)$: $a^Tb = 10 = \sqrt{20}\sqrt{5}$, equality, since a is a multiple of b.

SECTION 02

Matrices

We mostly distinguish square symmetric matrices from rectangular ones.

DEF · BASIC NOTIONS

- Trace: $\text{Tr}\{A\} = \sum_i A_{ii}$ for square A (sum of the diagonal). It is cyclic: $\text{Tr}\{AB\} = \text{Tr}\{BA\}$.
- Diagonals: $\text{diag}(A)$ extracts the diagonal of a matrix as a vector; $\text{Diag}(a)$ builds a diagonal matrix from a vector.
- Determinant: $\det(A)$.

Example: $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$: $\text{Tr}\,A = 5$, $\text{diag}(A) = (1, 4)$, $\text{Diag}(1, 4) = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$, $\det A = 4 - 6 = -2$.

PITFALL · CYCLIC, NOT COMMUTATIVE

$\mathrm{Tr}\{AB\} = \mathrm{Tr}\{BA\}$ even though $AB \neq BA$ in general. For three matrices you may *rotate*, $\mathrm{Tr}\{ABC\} = \mathrm{Tr}\{CAB\} = \mathrm{Tr}\{BCA\}$, but not swap: $\mathrm{Tr}\{ABC\} \neq \mathrm{Tr}\{BAC\}$ in general.

> 2.1 Eigenvalues (briefly)

DEF · EIGENPAIR

An eigenpair of a square matrix A is a scalar λ and a vector $v \neq 0$ with $Av = \lambda v$.

How to find eigenvalues. Rewrite $Av = \lambda v$ as $(A - \lambda I)v = 0$. A nonzero solution v exists exactly when $A - \lambda I$ is singular:

$$\det(A - \lambda I) = 0.$$

DEF · DIAGONALIZABLE

A is diagonalizable if it is similar to a diagonal matrix: $A = U\,\mathrm{Diag}\{\lambda_i\}U^{-1}$, with the eigenvectors as the columns of U.

EXAMPLE · NOT EVERY MATRIX IS DIAGONALIZABLE: $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

1. $\det(A - \lambda I) = \lambda^2$, so the only eigenvalue is $\lambda = 0$ (twice).
2. If $A = U\,\mathrm{Diag}(0,0)U^{-1}$, then $A = U \cdot 0 \cdot U^{-1} = 0$. But $A \neq 0$. Contradiction.

(The eigenvectors are only the multiples of $(1,0)$: there are not enough of them to build an invertible U.)

THEOREM 1.3: SPECTRAL THEOREM

Every **real symmetric** matrix has **real** eigenvalues and is diagonalizable by an **orthogonal** matrix:

$$A = U\,\mathrm{Diag}\{\lambda_i\}U^T, \qquad U^TU = UU^T = I.$$

WHY SYMMETRIC MATRICES ARE SO PLEASANT

For a symmetric matrix, U^{-1} is just U^T, and the eigenvectors are perpendicular. So a symmetric matrix only stretches space along perpendicular axes: by λ_1 along u_1, by λ_2 along u_2, and so on.

> 2.2 The 2×2 symmetric case

For $A = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$:

$$\det \begin{pmatrix} a - \lambda & c \\ c & b - \lambda \end{pmatrix} = (a - \lambda)(b - \lambda) - c^2 = \lambda^2 - (a + b)\lambda + (ab - c^2) = 0,$$

$$\lambda_{1,2} = \frac{a+b \pm \sqrt{(a+b)^2 - 4(ab-c^2)}}{2}.$$

TRACE AND DETERMINANT IN TERMS OF EIGENVALUES

For symmetric A (using cyclicity and $U^TU = I$):

$$\mathrm{Tr}\{A\} = \mathrm{Tr}\{U\,\mathrm{Diag}\{\lambda_i\}U^T\} = \mathrm{Tr}\{U^TU\,\mathrm{Diag}\{\lambda_i\}\} = \sum_i \lambda_i, \qquad \det(A) = \prod_i \lambda_i.$$

EXAMPLE · $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

$a = b = 2$, $c = 1$: $\lambda^2 - 4\lambda + 3 = 0$, so $\lambda \in \{3, 1\}$. Eigenvectors: $(A - 3I)v = 0 \Rightarrow v \propto (1, 1)$; $(A - I)v = 0 \Rightarrow v \propto (1, -1)$. Normalized:

$$U = \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad A = U\,\mathrm{Diag}(3, 1)U^T.$$

Check: $\mathrm{Tr}\,A = 4 = 3 + 1$ and $\det A = 4 - 1 = 3 = 3 \cdot 1$. The two eigenvectors are perpendicular: $(1, 1) \cdot (1, -1) = 0$.

SECTION 03

Positive definiteness

Positive definiteness is the matrix version of "$x > 0$". It is the most important matrix property in this course.

DEF · DEFINITION 1.4: PD AND PSD

A **symmetric** matrix X is

- positive definite, written $X \succ 0$, if $z^TXz > 0$ for all $z \neq 0$;
- positive semidefinite, written $X \succeq 0$, if $z^TXz \geq 0$ for all z.

For symmetric A, B we write $A \succ B$ to mean $A - B \succ 0$ (and $A \succeq B$ for $A - B \succeq 0$), just as with numbers.

EXAMPLE · COMPUTING z^TXz

$X = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $z = (z_1, z_2)$: $z^TXz = 2z_1^2 + 2z_1z_2 + 2z_2^2 = z_1^2 + z_2^2 + (z_1 + z_2)^2$. A sum of squares, and it is 0 only if $z_1 = z_2 = 0$. So $X \succ 0$.

$Y = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, $z = (1, -1)$: $z^TYz = 1 - 4 + 1 = -2 < 0$. So Y is *not* PSD, even though all its entries are positive.

PITFALL · POSITIVE ENTRIES ≠ POSITIVE DEFINITE

Y above has all entries positive and is not PSD. Conversely $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \succ 0$ has negative entries. PD is about $z^T X z$, not about the signs of the entries. Throughout the course, positivity statements always refer to **symmetric** matrices.

ONLY A PARTIAL ORDER

For numbers, either $a \geq b$ or $b \geq a$. For matrices this fails: two matrices may be incomparable. Example: $A = \mathrm{Diag}(2,1)$, $B = \mathrm{Diag}(1,2)$. Then $A - B = \mathrm{Diag}(1,-1)$ is neither $\succeq 0$ nor $\preceq 0$ ($z = (1,0)$ gives $+1$, $z = (0,1)$ gives -1). So neither $A \succeq B$ nor $B \succeq A$.

> 3.1 Two equivalent tests for $X \succ 0$

TEST 1: CHOLESKY

$X \succ 0$ **if and only if** $X = LL^T$ with L lower triangular with positive diagonal (the Cholesky decomposition, the matrix version of a square root). This is the standard *numerical* test for positive definiteness. For 2×2:

$$\begin{pmatrix} a & c \\ c & b \end{pmatrix} = \begin{pmatrix} \sqrt{a} & 0 \\ \frac{c}{\sqrt{a}} & \sqrt{b - \frac{c^2}{a}} \end{pmatrix} \begin{pmatrix} \sqrt{a} & \frac{c}{\sqrt{a}} \\ 0 & \sqrt{b - \frac{c^2}{a}} \end{pmatrix}, \qquad a > 0, \quad ab - c^2 > 0.$$

Why exactly those two conditions? The square roots must be of positive numbers: $a > 0$ for $\sqrt{a}$, and $b - c^2/a > 0$, i.e. (multiplying by $a > 0$) $ab - c^2 > 0$. If either fails, there is no real L with a positive diagonal.

EXAMPLE · 2×2 CHOLESKY

$A = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$: $a = 4$, $c = 2$, $b = 3$. $L_{11} = \sqrt{4} = 2$, $L_{21} = 2/2 = 1$, $L_{22} = \sqrt{3 - 4/4} = \sqrt{2}$.

$$L = \begin{pmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{pmatrix}, \qquad LL^T = \begin{pmatrix} 4 & 2 \\ 2 & 1+2 \end{pmatrix} = A. \checkmark$$

The construction generalizes recursively via the *Schur complement* (Chapter 8).

TEST 2: EIGENVALUES

$X \succ 0$ **if and only if** all eigenvalues of X are positive. ($X \succeq 0$ iff all are ≥ 0.)

Proof.

($\Rightarrow$) If $X = UDU^T \succ 0$, choose $z = u_j$ (the j-th eigenvector, a unit vector). Then $u_j^T X u_j = u_j^T(\lambda_j u_j) = \lambda_j > 0$.

($\Leftarrow$) If all $\lambda_i > 0$, then for any $z \neq 0$, set $w = U^T z \neq 0$ (U^T is invertible):

$$z^T X z = z^T U D U^T z = w^T D w = \sum_i \lambda_i w_i^2 = \sum_i \lambda_i [U^T z]_i^2 > 0.$$

2×2 SHORTCUT

Both eigenvalues of $\begin{pmatrix} a & c \\ c & b \end{pmatrix}$ are positive $\iff$ trace and determinant are positive:

$$a + b > 0 \text{ and } ab - c^2 > 0 \qquad \iff \qquad a > 0 \text{ and } ab - c^2 > 0.$$

Why the two versions agree. If $ab - c^2 > 0$ then $ab > c^2 \geq 0$, so a and b have the same sign. Then $a + b > 0$ and $a > 0$ say the same thing.

EXAMPLE · THREE QUICK CHECKS

X	a	$ab - c^2$	VERDICT
$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$	2	3	$\succ 0$ (eigenvalues 3, 1)
$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$	1	-3	not PSD (eigenvalues 3, -1)
$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$	1	0	$\succeq 0$ but not $\succ 0$ (eigenvalues 2, 0)

Food for thought (from the lecture). Does $A \succ B \succ 0$ imply $B^{-1} \succ A^{-1}$? For numbers: $3 > 2 > 0$ gives $\frac{1}{2} > \frac{1}{3}$. The matrix answer is **yes**; it is proved in the recitation, Exercise 1.8(2).

SECTION 04

Matrix norms

Matrices also have norms, but the terminology is less standard and can be confusing. Two matter here.

DEF · FROBENIUS NORM

$\|A\|_{\text{Fro}} = \sqrt{\sum_{ij} A_{ij}^2}$: treat A as one long vector and take its ℓ_2 norm.

DEF · SPECTRAL NORM

$\|A\|_2 = \max_{x:\, \|x\|_2 = 1} \|Ax\|_2$: the largest factor by which A can stretch a vector.

IN TERMS OF SINGULAR VALUES (CHAPTER 3)

The Frobenius norm is the ℓ_2 norm of the singular values; the spectral norm is their maximum. For a symmetric matrix the singular values are $|\lambda_i|$, so $\|A\|_2 = \max_i |\lambda_i|$.

EXAMPLE · BOTH NORMS OF $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

$\|A\|_{\text{Fro}} = \sqrt{4+1+1+4} = \sqrt{10} \approx 3.16$, and indeed $\sqrt{3^2+1^2} = \sqrt{10}$. $\|A\|_2 = \max(3,1) = 3$, attained at $x = \frac{1}{\sqrt{2}}(1,1)$: $Ax = \frac{1}{\sqrt{2}}(3,3)$, of length 3. Always $\|A\|_2 \le \|A\|_{\text{Fro}}$ (a maximum is at most the root of the sum of squares).

PITFALL · $\|A\|_2$ IS NOT THE ℓ_2 NORM OF THE ENTRIES

For a matrix, "$\|A\|_2$" means the *spectral* norm. The ℓ_2 norm of the entries is the Frobenius norm.

SECTION 05

Linear systems

The core computational problem of linear algebra is solving $Ax = b$. There are three possible cases:

- a **unique solution**, typically when A is square and invertible: $x = A^{-1}b$;
- **no solution**;
- **infinitely many solutions**.

EXAMPLE · ONE OF EACH

$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} x = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$: unique, $x = (1,1)$. $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} x = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$: none ($x_1 + x_2$ cannot be both 2 and 3). $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} x = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$: infinitely many ($x = (t, 2-t)$).

A FOURTH, IN-BETWEEN REGIME

In practice there is a fourth case: A is invertible but only *barely* (ill-conditioned). Then a tiny change in b can change x enormously. This is quantified by the condition number in the recitation (Exercise 1.4).

> 5.1 How do we solve linear systems numerically?

- **Direct (non-iterative) methods**: now and in the recitation. They give the exact answer (up to rounding) in a fixed number of steps.
- **Iterative methods**: in a few weeks.

The two classical direct methods are the LU decomposition (general square matrices, Part C) and the Cholesky decomposition (positive definite matrices, below). Both solve systems in a numerically stable and efficient way.

PITFALL · DON'T COMPUTE A^{-1}

$x = A^{-1}b$ is the right formula on paper, but numerically we never form A^{-1}. It costs more and is less stable. We factor A into triangular pieces and solve triangular systems instead [BV C.2].

SECTION 06

Solving with Cholesky

If $A \in \mathbb{R}^{n\times n}$ is symmetric positive definite, it has a Cholesky decomposition $A = LL^T$, where L is the *unique* lower triangular matrix with positive diagonal.

COMPUTING L (GOLUB & VAN LOAN, ALG. 4.2.1)

Initialize L as an $n \times n$ zero matrix.
for $j = 1, \dots, n$:
  $v_{j:n} = A_{j:n,\,j}$ (the part of column j on and below the diagonal)
  for $k = 1, \dots, j-1$: $v_{j:n} = v_{j:n} - L_{j,k}\, L_{j:n,\,k}$ (remove earlier columns)
  $L_{j:n,\,j} = v_{j:n}/\sqrt{v_j}$ (normalize by the root of the new pivot)
After the loop, L is lower triangular and $A = LL^T$.

SOLVING $Ax = b$

1. Compute L with $A = LL^T$.
2. Solve $Ly = b$ by forward substitution (top row first).
3. Solve $L^T x = y$ by backward substitution (bottom row first).

Why it works: $Ax = L(L^T x) = Ly = b$.

EXAMPLE · A FULL 3×3 SOLVE

$A = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 6 \end{pmatrix}$, $b = (10, 13, 17)$.

Factor.

- $j = 1$: $v = (4, 2, 2)$, $L_{:,1} = v/\sqrt{4} = (2, 1, 1)$.
- $j = 2$: $v_{2:3} = (5, 3) - L_{21}\, L_{2:3,1} = (5, 3) - 1 \cdot (1, 1) = (4, 2)$; $L_{2:3,2} = (4, 2)/\sqrt{4} = (2, 1)$.
- $j = 3$: $v_3 = 6 - L_{31}L_{31} - L_{32}L_{32} = 6 - 1 - 1 = 4$; $L_{33} = \sqrt{4} = 2$.

$L = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 2 \end{pmatrix}$. All pivots were positive, which also certifies $A \succ 0$.

Forward $Ly = b$: $y_1 = 10/2 = 5$, $y_2 = (13 - 1 \cdot 5)/2 = 4$, $y_3 = (17 - 5 - 4)/2 = 4$.

Backward $L^T x = y$ with $L^T = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$: $x_3 = 4/2 = 2$, $x_2 = (4-2)/2 = 1$, $x_1 = (5-1-2)/2 = 1$.

$x = (1, 1, 2)$. Check: $Ax = (4+2+4,\ 2+5+6,\ 2+3+12) = (10, 13, 17) = b$. ✓

COMPLEXITY

Cholesky factorization $\approx \frac{1}{3}n^3$ flops; each triangular solve $O(n^2)$; total $O(n^3)$, but with half the constant of LU ($\approx \frac{2}{3}n^3$).

EXAMPLE · WHAT THE CONSTANTS MEAN

For $n = 1000$: Cholesky $\approx 3.3 \times 10^8$ flops, LU $\approx 6.7 \times 10^8$, one triangular solve $\approx 10^6$. The factorization dominates. Once it is done, every new right-hand side b is cheap.

PART B · RECITATION 1

Exercises: Norms, Conditioning, PSD Matrices

SECTION 07

Exercise 1.1: three norms of three vectors

EXAMPLE · EXERCISE 1.1 ●○○

Compute $\|x\|_1, \|x\|_2, \|x\|_\infty$ for $x = (1,0,0,0)$, $(1,1,1,1)$, $(3,-1,2,0)$ and verify $\|x\|_\infty \le \|x\|_2 \le \|x\|_1$. Which vectors make which inequalities tight?

Solution.

x	$\|x\|_1$	$\|x\|_2$	$\|x\|_\infty$	COMMENT
$(1,0,0,0)$	1	1	1	1-sparse: all norms coincide
$(1,1,1,1)$	4	$\sqrt{4} = 2$	1	flat: as far apart as possible
$(3,-1,2,0)$	$3+1+2=6$	$\sqrt{14} \approx 3.74$	3	$3 < 3.74 < 6$ (strict)

SPARSE VS. FLAT

A vector with a single nonzero entry collapses all three norms to the same value. A flat vector spreads them as far apart as possible. For $(1,1,1,1)$: $\|x\|_2 = 2 = \sqrt{4}\cdot\|x\|_\infty$ and $\|x\|_1 = 4 = \sqrt{4}\cdot\|x\|_2$, so the upper constants of Exercise 1.3 are attained.

SECTION 08

Exercise 1.2: ℓ_2 is a norm

EXAMPLE · EXERCISE 1.2 ●○○

Using only the properties of the inner product and Cauchy–Schwarz, verify that $f(x) = \sqrt{x^T x}$ satisfies the three norm axioms.

Solution.

1. **Positive definiteness.** $x^T x = \sum_i x_i^2 \ge 0$ (a sum of squares), and it is 0 iff every $x_i = 0$. The square root keeps both facts.
2. **Homogeneity.** $\sqrt{(\alpha x)^T(\alpha x)} = \sqrt{\alpha^2\, x^T x} = \sqrt{\alpha^2}\sqrt{x^T x} = |\alpha|\sqrt{x^T x}$.
3. **Triangle inequality.** Expand the square, then use Cauchy–Schwarz on the middle term

($x^T y \le \|x\|_2\|y\|_2$):

$$\|x+y\|_2^2 = x^Tx + 2x^Ty + y^Ty \le \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2 = (\|x\|_2 + \|y\|_2)^2.$$

Both sides are nonnegative, so take square roots.

PITFALL · LOGICAL ORDER

Cauchy–Schwarz was proved directly (lecture), and the triangle inequality *follows* from it, not the other way around. Using the triangle inequality for ℓ_2 inside this proof would be circular.

PITFALL · $\sqrt{\alpha^2} = |\alpha|$

Not α. With $\alpha = -2$: $\sqrt{4} = 2 = |-2|$. This is where the absolute value in the homogeneity axiom comes from.

SECTION 09

Exercise 1.3: sharp equivalence constants

EXAMPLE · EXERCISE 1.3 ●●○

Prove for all $x \in \mathbb{R}^n$: $\|x\|_2 \le \|x\|_1 \le \sqrt{n}\,\|x\|_2$ and $\|x\|_\infty \le \|x\|_2 \le \sqrt{n}\,\|x\|_\infty$, and exhibit vectors attaining each inequality with equality.

Solution, one inequality at a time.

1. $\|x\|_\infty \le \|x\|_2$: for every i, $x_i^2 \le \sum_j x_j^2$, so $|x_i| \le \|x\|_2$. Take the max over i.
2. $\|x\|_2 \le \sqrt{n}\|x\|_\infty$: replace every entry by the largest one, $\|x\|_2^2 = \sum_i x_i^2 \le \sum_i \left(\max_j |x_j|\right)^2 = n\|x\|_\infty^2$. Take square roots.
3. $\|x\|_2 \le \|x\|_1$: $\left(\sum_i |x_i|\right)^2 = \sum_i x_i^2 + \sum_{i\neq j} |x_i||x_j| \ge \sum_i x_i^2$ (the cross terms are ≥ 0). Take square roots.
4. $\|x\|_1 \le \sqrt{n}\|x\|_2$: Cauchy–Schwarz with the all-ones vector $\mathbf{1}$ and $|x| = (|x_1|, \dots, |x_n|)$:

$$\|x\|_1 = \mathbf{1}^T|x| \le \|\mathbf{1}\|_2\,\||x|\|_2 = \sqrt{n}\,\|x\|_2.$$

Tightness. The left inequalities are equalities for 1-sparse vectors such as $(1, 0, \dots, 0)$ (all norms equal 1). The right ones are equalities for flat vectors such as $\mathbf{1} = (1, \dots, 1)$: $\|\mathbf{1}\|_2 = \sqrt{n} = \sqrt{n}\|\mathbf{1}\|_\infty$ and $\|\mathbf{1}\|_1 = n = \sqrt{n} \cdot \sqrt{n}$ (equality in Cauchy–Schwarz, since $|x|$ is a multiple of $\mathbf{1}$).

SECTION 10

Exercise 1.4: the condition number

DEF · CONDITION NUMBER

For invertible $A \in \mathbb{R}^{n\times n}$: $\kappa(A) = \|A\|_2 \cdot \|A^{-1}\|_2$. For symmetric A this is $\dfrac{\max_i |\lambda_i|}{\min_i |\lambda_i|}$. Always $\kappa(A) \geq 1$.

EXAMPLE · EXERCISE 1.4 ●●○

(1) Prove: if $Ax = b$ and $A\tilde{x} = b + \delta b$, then $\dfrac{\|\tilde{x} - x\|_2}{\|x\|_2} \leq \kappa(A) \cdot \dfrac{\|\delta b\|_2}{\|b\|_2}$.

(2) Demonstrate it on $A = \begin{pmatrix} 1 & 0.99 \\ 0.99 & 0.98 \end{pmatrix}$, $b = \begin{pmatrix} 0.99 \\ 0.98 \end{pmatrix}$, $\tilde{b} = \begin{pmatrix} 0.99 \\ 0.981 \end{pmatrix}$.

The tool used twice: $\|Mv\|_2 \leq \|M\|_2\|v\|_2$. It follows from the definition of the spectral norm: for $v \neq 0$, $u = v/\|v\|$ is a unit vector, and a maximum is at least each of its values, so $\|Mv\|/\|v\| = \|Mu\| \leq \|M\|_2$.

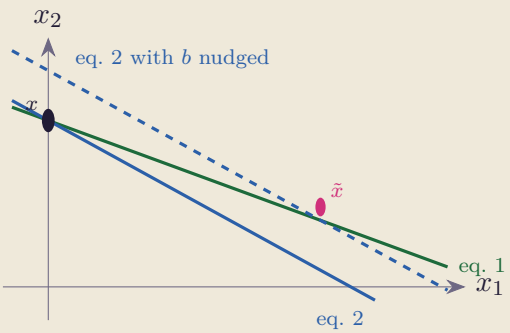
Solution (1).

1. Subtract the two systems: $A\tilde{x} - Ax = \delta b$, i.e. $A(\tilde{x} - x) = \delta b$, so $\tilde{x} - x = A^{-1}\delta b$.
2. Apply the tool with $M = A^{-1}$: $\|\tilde{x} - x\|_2 = \|A^{-1}\delta b\|_2 \leq \|A^{-1}\|_2 \, \|\delta b\|_2$.
3. Apply it with $M = A$, $v = x$: $\|b\|_2 = \|Ax\|_2 \leq \|A\|_2\|x\|_2$, hence $\dfrac{1}{\|x\|_2} \leq \dfrac{\|A\|_2}{\|b\|_2}$ (assume $b \neq 0$).
4. Multiply the inequalities of steps 2 and 3 (all terms nonnegative):

$$\frac{\|\tilde{x} - x\|_2}{\|x\|_2} \leq \underbrace{\|A\|_2\|A^{-1}\|_2}_{\kappa(A)} \frac{\|\delta b\|_2}{\|b\|_2}.$$

Solution (2), with numbers.

1. $\det A = 1 \cdot 0.98 - 0.99^2 = 0.98 - 0.9801 = -10^{-4}$: invertible, but barely.
2. Eigenvalues ($\mathrm{Tr} = 1.98$, $\det = -10^{-4}$): $\lambda = \frac{1.98 \pm \sqrt{1.98^2 + 4\cdot 10^{-4}}}{2}$, so $\lambda_1 \approx 1.98$, $\lambda_2 \approx -5.05 \cdot 10^{-5}$. A is symmetric, so $\kappa(A) = |\lambda_1|/|\lambda_2| \approx 3.9 \cdot 10^4$.
3. Solve $Ax = b$: b is exactly the second column of A, so $x = (0, 1)$.
4. Solve $A\tilde{x} = \tilde{b}$ (Cramer's rule): $\tilde{x}_1 = \frac{0.99\cdot 0.98 - 0.99\cdot 0.981}{-10^{-4}} = \frac{-0.00099}{-10^{-4}} = 9.9$, $\tilde{x}_2 = \frac{1\cdot 0.981 - 0.99\cdot 0.99}{-10^{-4}} = \frac{0.0009}{-10^{-4}} = -9$.
   Check: $9.9 - 0.99 \cdot 9 = 0.99$ and $0.99 \cdot 9.9 - 0.98 \cdot 9 = 9.801 - 8.82 = 0.981$. ✓
5. Relative changes: $\dfrac{\|\tilde{x} - x\|}{\|x\|} = \|(9.9, -10)\| = \sqrt{198.01} \approx 14.1$, while $\dfrac{\|\tilde{b} - b\|}{\|b\|} = \dfrac{0.001}{\sqrt{0.9801 + 0.9604}} \approx 7.2 \cdot 10^{-4}$.
6. Amplification $\approx 14.1/(7.2 \cdot 10^{-4}) \approx 2 \cdot 10^4 \leq \kappa \approx 3.9 \cdot 10^4$. Same order, consistent with the bound.



Schematic: the two equations of an ill-conditioned system are almost parallel lines. Moving one line a tiny bit slides their crossing point far along them. That is why $\tilde{x}$ is so far from x.

MORALS

- $\kappa(A) \approx 1$: well-conditioned. $\kappa(A) \gg 1$: small errors in b (or rounding errors) may cause large errors in x.
- The condition number is a property of the matrix, not of the algorithm. No clever solver can undo it.

SECTION 11

Exercise 1.5: the matrix inner product

EXAMPLE · EXERCISE 1.5 ●○○

For $A, B \in \mathbb{R}^{m\times n}$ define $\langle A, B\rangle = \mathrm{Tr}\{B^T A\}$. (1) Show it is an inner product on $\mathbb{R}^{m\times n}$. Deduce (2) $\|A\|_{\mathrm{Fro}}^2 = \mathrm{Tr}\{AA^T\}$, (3) $\mathrm{Tr}\{AB^T\} = \mathrm{Tr}\{A^T B\}$, (4) $\mathrm{Tr}\{xy^T\} = x^T y$ for vectors x, y.

Solution.

1. Write it with indices: $(B^T A)_{jj} = \sum_i B_{ij}A_{ij}$, so $\mathrm{Tr}\{B^T A\} = \sum_{ij} A_{ij}B_{ij}$. Stack the entries of each matrix into a long vector in $\mathbb{R}^{mn}$: this is exactly the standard inner product there. So symmetry, linearity and positive definiteness hold because they hold in $\mathbb{R}^{mn}$.
2. Take $B = A$: $\langle A, A\rangle = \sum_{ij} A_{ij}^2 = \|A\|_{\mathrm{Fro}}^2$ (the Frobenius norm is the norm induced by this inner product). By cyclicity, $\mathrm{Tr}\{A^T A\} = \mathrm{Tr}\{AA^T\}$.
3. Both sides equal $\sum_{ij} A_{ij}B_{ij}$.
4. A special case of (3) with $m \times 1$ matrices, or directly: $\mathrm{Tr}\{xy^T\} = \mathrm{Tr}\{y^T x\} = y^T x = x^T y$, since the trace of a scalar is itself. This little identity is used constantly from Chapter 9 on.

EXAMPLE · NUMBERS

$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: $\langle A, B\rangle = 1\cdot 0 + 2\cdot 1 + 3\cdot 1 + 4\cdot 0 = 5$. Via the trace: $B^T A = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$, trace $3 + 2 = 5$. ✓

SECTION 12

Exercise 1.6: entries of a PSD matrix

DEF · THE THREE MATRIX SETS

$$S^d = \{A \in \mathbb{R}^{d\times d} : A = A^T\}, \qquad S^d_+ = \{A \in S^d : A \succeq 0\}, \qquad S^d_{++} = \{A \in S^d : A \succ 0\}.$$

Symmetric, positive semidefinite, and positive definite matrices.

EXAMPLE · EXERCISE 1.6 ●○○

Let $A \in S^d_+$. Prove: (1) $A_{ii} \geq 0$ for every i; (2) $\det(A) \geq 0$; (3) the maximal entry of A is on the diagonal, $\max_{l,k} A_{lk} = \max_i A_{ii}$; (4) if $A_{ii} = 0$ for some i, then $A_{ij} = 0$ for every j.

Solution. The trick throughout: choose a clever test vector z and use $z^T A z \geq 0$.

1. With the standard basis vector $z = e_i$: $A_{ii} = e_i^T A e_i \geq 0$.
2. By the spectral theorem $A = U\Sigma U^T$, so $\det(A) = \det(U)\det(\Sigma)\det(U^T) = \det(UU^T)\det(\Sigma) = \det(I)\prod_i \Sigma_{ii} = \prod_i \Sigma_{ii}$. All eigenvalues of a PSD matrix are ≥ 0, so $\det(A) \geq 0$.
3. Suppose, toward a contradiction, that the maximum is A_{ij} with $i \neq j$ and A_{ij} is strictly larger than every diagonal entry. Take $z = e_i - e_j$:

$$(e_i - e_j)^T A(e_i - e_j) = A_{ii} - 2A_{ij} + A_{jj} = (A_{ii} - A_{ij}) + (A_{jj} - A_{ij}) < 0,$$

since both brackets are negative. This contradicts $A \succeq 0$.

4. Suppose $A_{ii} = 0$ but $A_{ij} \neq 0$ for some j. Look at the 2×2 principal submatrix $\tilde{A} = \begin{pmatrix} A_{ii} & A_{ij} \\ A_{ji} & A_{jj} \end{pmatrix} = \begin{pmatrix} 0 & A_{ij} \\ A_{ij} & A_{jj} \end{pmatrix}$. Its determinant is $-A_{ij}^2 < 0$, so (by part 2) $\tilde{A}$ is not PSD: some $x = (x_1, x_2)$ has $x^T \tilde{A} x < 0$. Then

$$(x_1 e_i + x_2 e_j)^T A(x_1 e_i + x_2 e_j) = x^T \tilde{A} x < 0,$$

contradicting $A \succeq 0$.

EXAMPLE · SEEING (3) AND (4) FAIL

$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$: the largest entry 2 is off the diagonal, so by (3) it cannot be PSD (indeed $z = (1, -1)$ gives -2). $\begin{pmatrix} 0 & 1 \\ 1 & 5 \end{pmatrix}$: a zero on the diagonal with a nonzero entry in its row, so by (4) it cannot be PSD (determinant -1).

SECTION 13

LU in the notes and Exercise 1.7

The notes' recitation summarises LU decomposition ($PA = LU$, Gaussian elimination, partial pivoting, cost). Part C explains it in full, with worked 3×3 examples. Exercise 1.7 counts the cost of the triangular solves used there.

EXAMPLE · EXERCISE 1.7 ●○○

Count the flops of solving $Ly = b$ by forward substitution (L lower triangular, $n \times n$).

Solution. Row i reads $L_{i1}y_1 + \cdots + L_{i,i-1}y_{i-1} + L_{ii}y_i = b_i$, so $y_i = (b_i - \sum_{k<i} L_{ik}y_k)/L_{ii}$. That is $i-1$ multiplications, $i-1$ subtractions and one division: $2i-1$ flops. Summing,

$$\sum_{i=1}^{n}(2i-1) = 2 \cdot \frac{n(n+1)}{2} - n = n^2.$$

EXAMPLE · CHECK FOR $n = 3$

Rows cost $1, 3, 5$ flops: total $9 = 3^2$. ✓ (With L's diagonal equal to 1, as in LU, the divisions disappear and the count is $n^2 - n$; still $\approx n^2$.)

SECTION 14

Exercise 1.8: the PSD order (Moed C 2024)

EXAMPLE · EXERCISE 1.8 ●●●

Let X, Y be symmetric. (1) If $X \succeq 0$ and $Y \succeq 0$, then $\mathrm{Tr}\{XY\} \geq 0$. (2) If $X \succeq Y \succ 0$, then $X^{-1} \preceq Y^{-1}$ (this answers the lecture's food-for-thought question). (3) If $X \succeq Y \succeq 0$, then $MXM^T \succeq MYM^T$ for any matrix M (not necessarily square).

We do part (3) first, because part (2) uses it.

Solution (3). For any z of the right size, put $w = M^T z$. Using $X \succeq Y$ with the vector w:

$$z^T MXM^T z = w^T X w \geq w^T Y w = z^T MYM^T z.$$

So $z^T(MXM^T - MYM^T)z \geq 0$ for all z, which is the claim.

Solution (1). Factor $Y = UU^T$ (e.g. Cholesky, or $U = Y^{1/2}$). By cyclicity of the trace,

$$\mathrm{Tr}\{XY\} = \mathrm{Tr}\{XUU^T\} = \mathrm{Tr}\{U^T XU\} = \sum_i u_i^T X u_i \geq 0,$$

where u_i are the columns of U (the diagonal entries of $U^T XU$ are $u_i^T X u_i$), and each term is ≥ 0 because $X \succeq 0$.

Solution (2).

1. Apply part (3) with $M = Y^{-1/2}$ (symmetric, so $M^T = M$): $Y^{-1/2}XY^{-1/2} \succeq Y^{-1/2}YY^{-1/2} = I$.
2. A symmetric matrix $\succeq I$ has all eigenvalues ≥ 1, so its inverse has all eigenvalues ≤ 1: $(Y^{-1/2}XY^{-1/2})^{-1} = Y^{1/2}X^{-1}Y^{1/2} \preceq I$.

3. Apply part (3) again with $M = Y^{-1/2}$: $X^{-1} \preceq Y^{-1/2}\, I\, Y^{-1/2} = Y^{-1}$.

EXAMPLE · CHECKING (2) ON DIAGONAL MATRICES

$X = \mathrm{Diag}(4, 2) \succeq Y = \mathrm{Diag}(1, 2) \succ 0$ (since $X - Y = \mathrm{Diag}(3, 0) \succeq 0$). Then $X^{-1} = \mathrm{Diag}(\frac{1}{4}, \frac{1}{2})$ and $Y^{-1} = \mathrm{Diag}(1, \frac{1}{2})$, and $Y^{-1} - X^{-1} = \mathrm{Diag}(\frac{3}{4}, 0) \succeq 0$. ✓ For diagonal matrices this is just "$a \geq b > 0 \Rightarrow \frac{1}{a} \leq \frac{1}{b}$" entry by entry.

PITFALL · ORDER OF THE SOLUTION

The notes prove (2) using (3) before (3) is proved. That is fine logically, since (3) does not depend on (2), but read (3) first.

PART C · RECITATION 1 HANDOUT

LU Decomposition and Solving $Ax = b$

SECTION 15

What do we want?

Given a square invertible $n \times n$ matrix A and a vector b, find x with $Ax = b$. We could compute A^{-1} and multiply, but that is expensive and numerically less stable. **The idea of LU:** factor A *once* into two triangular matrices. After that, every solve (for every new b) is very cheap.

SECTION 16

Why are triangular systems easy?

Lower triangular $Ly = b$. The first row contains only y_1, the second only y_1, y_2, and so on. So we solve top to bottom: forward substitution.

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix} \quad \Rightarrow \quad y_1 = 4, \quad y_2 = 10 - 2 \cdot 4 = 2, \quad y_3 = 24 - 4 \cdot 4 - 3 \cdot 2 = 2.$$

Upper triangular $Ux = y$. Start from the last row and go up: backward substitution.

Each triangular system costs about n^2 operations (Exercise 1.7), compared with about n^3 for a general system.

SECTION 17

The idea: LU is exactly Gaussian elimination

Row-reduce A as in Linear Algebra 1: subtract a multiple of the pivot row from every row below it, until A becomes upper triangular. Call the result U. Store the multipliers you used in a matrix L (lower triangular, 1s on the diagonal), exactly in the position where you created the zero.

LU DECOMPOSITION

$$A = LU$$

- L: lower triangular, with 1 on the diagonal; it holds the **multipliers**.
- U: upper triangular; it is the **result of the elimination**.

Notation: R_i = row i. The operation $R_i \leftarrow R_i - mR_k$ means: subtract m times row k from

row i.

SECTION 18

The algorithm (without row swaps)

LU BY GAUSSIAN ELIMINATION

$U \leftarrow A,\ L \leftarrow I$
for $k = 1, \dots, n-1$ (column k; the pivot is U_{kk})
  for $i = k+1, \dots, n$ (every row below the pivot)
    $L_{ik} \leftarrow U_{ik}/U_{kk}$ (the multiplier)
    $R_i(U) \leftarrow R_i(U) - L_{ik}R_k(U)$ (now $U_{ik} = 0$)
result: $A = LU$

In words: go column by column. In column k the pivot is the diagonal entry U_{kk}. For each row below it compute a multiplier (the entry divided by the pivot), write it into L, and zero the entry by a row subtraction.

SECTION 19

Example 1: factoring a 3×3 matrix

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix}.$$

Step $k = 1$ (pivot $U_{11} = 2$):

- Row 2: multiplier $4/2 = 2$, so $R_2 \leftarrow R_2 - 2R_1 = (4,3,3) - 2 \cdot (2,1,1) = (0,1,1)$.
- Row 3: multiplier $8/2 = 4$, so $R_3 \leftarrow R_3 - 4R_1 = (8,7,9) - 4 \cdot (2,1,1) = (0,3,5)$.

After the first step ($*$ = not yet computed):

$$U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 5 \end{pmatrix}, \qquad L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & * & 1 \end{pmatrix}.$$

Step $k = 2$ (pivot $U_{22} = 1$):

- Row 3: multiplier $3/1 = 3$, so $R_3 \leftarrow R_3 - 3R_2 = (0,3,5) - 3 \cdot (0,1,1) = (0,0,2)$.

Result:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}, \qquad A = LU.$$

Check. Each row of LU is a combination of the rows of U with the multipliers:

- Row 1: $(2, 1, 1)$.
- Row 2: $2 \cdot (2, 1, 1) + (0, 1, 1) = (4, 3, 3)$.
- Row 3: $4 \cdot (2, 1, 1) + 3 \cdot (0, 1, 1) + (0, 0, 2) = (8, 7, 9)$.

This is exactly A. In other words, L undoes the elimination and gives back A.

SECTION 20

Solving $Ax = b$ with the factorization

$$Ax = b \iff L\underbrace{(Ux)}_{y} = b.$$

Step A: solve $Ly = b$ (forward substitution). **Step B:** solve $Ux = y$ (backward substitution).

EXAMPLE · $b = (4, 10, 24)$

Step A, $Ly = b$, top to bottom: $y_1 = 4$, $y_2 = 10 - 2 \cdot 4 = 2$, $y_3 = 24 - 4 \cdot 4 - 3 \cdot 2 = 2$, so $y = (4, 2, 2)$.

Step B, $Ux = y$, bottom to top:

$$2x_3 = 2 \Rightarrow x_3 = 1, \qquad x_2 + x_3 = 2 \Rightarrow x_2 = 1, \qquad 2x_1 + x_2 + x_3 = 4 \Rightarrow x_1 = 1.$$

So $x = (1, 1, 1)$. Check: $A(1, 1, 1)^T = (2 + 1 + 1,\ 4 + 3 + 3,\ 8 + 7 + 9) = (4, 10, 24) = b$. ✓

WHAT y REALLY IS

y is exactly what would appear in the b column if you row-reduced the augmented matrix $[A \,|\, b]$. The factorization stores the elimination, and every new b just goes through the same operations.

SECTION 21

Why do we need row swaps (P)?

At every step we divide by the pivot U_{kk}. Two things can go wrong:

1. **A zero pivot**: we cannot divide. For example $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$: elimination cannot even start.
2. **A tiny pivot**: the multipliers are huge, and the computer's rounding errors destroy the answer.

EXAMPLE · EXAMPLE OF THE PROBLEM: A TINY PIVOT

$A = \begin{pmatrix} \varepsilon & 1 \\ 1 & 1 \end{pmatrix}$, $b = (1, 2)$, with $\varepsilon = 10^{-20}$. The true solution is about $x = (1, 1)$.

- **Without a swap**: the multiplier is $1/\varepsilon = 10^{20}$. On a computer $1 - 10^{20}$ rounds to -10^{20} and $2 - 10^{20}$ also rounds to -10^{20}. This gives $x_2 = 1$ and then $x_1 = (1 - 1)/\varepsilon = 0$. **Wrong.**
- **With a swap**: swap the rows to get $\begin{pmatrix} 1 & 1 \\ \varepsilon & 1 \end{pmatrix}$. The multiplier is ε (small and safe), and we get $x = (1, 1)$. **Correct.**

PARTIAL PIVOTING

Before zeroing column k, swap rows so that the pivot is the entry of largest absolute value in that column (from row k down).

SECTION 22

What is the matrix P?

DEF · PERMUTATION MATRIX

P is a permutation matrix: the identity matrix with its rows swapped. Multiplying PA swaps the rows of A in the same way. For example $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ swaps rows 2 and 3.

Each row of A is an equation, and changing the order of the equations does not change the solution x. We only have to reorder b in the same way. The result of elimination with swaps is

$$PA = LU.$$

A SIMPLE WAY TO THINK ABOUT IT

If we knew the right order of the equations in advance, we would set $A' = PA$ and run ordinary LU with no swaps. The problem is that a bad pivot only shows up *during* the elimination (in the next example the pivot becomes 0 only after the first step). So we swap as we go, and record the swaps in P.

SECTION 23

The algorithm with swaps

LU WITH PARTIAL PIVOTING

$U \leftarrow A,\ L \leftarrow I,\ P \leftarrow I$
for $k = 1, \dots, n-1$
  $r \leftarrow$ the row $i \geq k$ with the largest $|U_{ik}|$ (choose the pivot)
  swap rows k, r in U
  swap rows k, r in P (remember the swap)
  swap the multipliers already in L: $L_{k,1:k-1} \leftrightarrow L_{r,1:k-1}$
  for $i = k+1, \dots, n$: $L_{ik} \leftarrow U_{ik}/U_{kk},\ R_i(U) \leftarrow R_i(U) - L_{ik}R_k(U)$
result: $PA = LU$

WHY SWAP THE MULTIPLIERS IN L TOO?

Every multiplier belongs to a particular row. If the row moves, its multiplier moves with it. That way the result is identical to ordinary LU on PA.

SECTION 24

Example 2: factoring with a row swap

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 5 \\ 4 & 6 & 8 \end{pmatrix}.$$

Start with $U = A$, $L = I$, $P = I$. (On the board we swap only when we must, to keep the numbers simple; see the note at the end.)

Step $k = 1$ (pivot 1):

- Row 2: multiplier 2, so $R_2 \leftarrow R_2 - 2R_1 = (0, 0, 3)$.
- Row 3: multiplier 4, so $R_3 \leftarrow R_3 - 4R_1 = (0, 2, 4)$.

$$U = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 3 \\ 0 & 2 & 4 \end{pmatrix}, \qquad L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & * & 1 \end{pmatrix}.$$

Step $k = 2$**:** the pivot is $U_{22} = 0$. We cannot divide! Swap rows 2 and 3:

- in U: $U = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{pmatrix}$;
- in P: $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$;
- in the multipliers already in L (column 1): $2 \leftrightarrow 4$, so now $L_{21} = 4$ and $L_{31} = 2$.

Continue: the multiplier for row 3 is $0/2 = 0$, so U does not change and $L_{32} = 0$.

Result:

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad L = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{pmatrix}, \qquad PA = LU.$$

Check. PA is A with rows 2 and 3 swapped: $PA = \begin{pmatrix} 1 & 1 & 1 \\ 4 & 6 & 8 \\ 2 & 2 & 5 \end{pmatrix}$. On the other side:

- Row 1 of LU: $(1, 1, 1)$.
- Row 2 of LU: $4 \cdot (1, 1, 1) + (0, 2, 4) = (4, 6, 8)$.
- Row 3 of LU: $2 \cdot (1, 1, 1) + 0 \cdot (0, 2, 4) + (0, 0, 3) = (2, 2, 5)$.

Exactly PA. (And ordinary LU without swaps on $A' = PA$ gives exactly the same L and U.)

Note: what the computer actually does. Libraries such as LAPACK and numpy/scipy always swap to the largest entry, here already at the first step (the entry 4). The result is then: P swaps rows 1 and 3,

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 1/4 & 1/2 & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 4 & 6 & 8 \\ 0 & -1 & 1 \\ 0 & 0 & -3/2 \end{pmatrix}.$$

On the board we swap only when we must, so that the numbers stay simple. The idea is the same.

SECTION 25

Solving $Ax = b$ with P

$$Ax = b \iff PAx = Pb \iff L\underbrace{(Ux)}_{y} = Pb.$$

1. Factor once: $PA = LU$.
2. For each b: reorder it, Pb (only a change of order, no arithmetic).
3. Solve $Ly = Pb$ (forward substitution).
4. Solve $Ux = y$ (backward substitution).

EXAMPLE · $b = (3, 9, 18)$

Step 2: swap entries 2 and 3: $Pb = (3, 18, 9)$.

Step 3, $Ly = Pb$: $y_1 = 3$, $y_2 = 18 - 4 \cdot 3 = 6$, $y_3 = 9 - 2 \cdot 3 - 0 \cdot 6 = 3$, so $y = (3, 6, 3)$.

Step 4, $Ux = y$:

$$3x_3 = 3 \Rightarrow x_3 = 1, \qquad 2x_2 + 4x_3 = 6 \Rightarrow x_2 = 1, \qquad x_1 + x_2 + x_3 = 3 \Rightarrow x_1 = 1.$$

So $x = (1, 1, 1)$. Check: $A(1, 1, 1)^T = (1 + 1 + 1,\ 2 + 2 + 5,\ 4 + 6 + 8) = (3, 9, 18) = b$. ✓

PITFALL · FORGETTING P

If you solve $Ly = b$ instead of $Ly = Pb$, you get $y = (3, -3, 12)$ and then $x = (8.5, -9.5, 4)$. **Wrong.** (It is in fact the solution of a different system, $Ax = Pb$.)

SECTION 26

Summary of LU

ONE SENTENCE TO REMEMBER

$PA = LU$ **is elimination with memory**: U = the result of the elimination, L = the multipliers, P = the order of the rows.

STEP	COST (FLOPS, LARGE n)
Factor $PA = LU$ (once)	$\approx \frac{2}{3}n^3$
Reorder Pb	0
$Ly = Pb$ (forward substitution)	$\approx n^2$
$Ux = y$ (backward substitution)	$\approx n^2$

So if you need to solve for many different b's, factor once, and each further solve costs only about $2n^2$.

COMPARISON WITH CHOLESKY

If A is symmetric positive definite, we can write $A = LL^T$, i.e. $U = L^T$. That is half the work ($\approx \frac{1}{3}n^3$), and no P is needed: the pivots are always positive. LU is the general case, for any square invertible matrix.

WEEK 01

Practice Problems and Summary

SECTION 27

Practice problems with full solutions

EXAMPLE · P1: NORMS

$x = (2, -2, 1)$. Compute $\|x\|_1, \|x\|_2, \|x\|_\infty, \|x\|_0$ and check both chains of Exercise 1.3 with $n = 3$.

Solution. $\|x\|_1 = 5$, $\|x\|_2 = \sqrt{4+4+1} = 3$, $\|x\|_\infty = 2$, $\|x\|_0 = 3$. $2 \le 3 \le \sqrt{3} \cdot 2 \approx 3.46$ ✓ and $3 \le 5 \le \sqrt{3} \cdot 3 \approx 5.20$ ✓.

EXAMPLE · P2: POSITIVE DEFINITE OR NOT?

For which c is $X = \begin{pmatrix} 2 & c \\ c & 8 \end{pmatrix} \succ 0$? For $c = 3$, find the Cholesky factor.

Solution. $a = 2 > 0$, so we need $ab - c^2 = 16 - c^2 > 0$, i.e. $|c| < 4$. For $c = 3$: $L_{11} = \sqrt{2}$, $L_{21} = 3/\sqrt{2}$, $L_{22} = \sqrt{8 - 9/2} = \sqrt{3.5}$. Check: $L_{21}^2 + L_{22}^2 = 4.5 + 3.5 = 8$. ✓

EXAMPLE · P3: EIGEN-DECOMPOSITION AND NORMS

$A = \begin{pmatrix} 5 & 0 \\ 0 & -2 \end{pmatrix}$. Find $\|A\|_2$, $\|A\|_{\text{Fro}}$, $\kappa(A)$. Is $A \succeq 0$?

Solution. Diagonal and symmetric, eigenvalues $5, -2$. $\|A\|_2 = \max(5, 2) = 5$, $\|A\|_{\text{Fro}} = \sqrt{25 + 4} = \sqrt{29} \approx 5.39$, $\kappa = 5/2 = 2.5$. Not PSD ($e_2^T A e_2 = -2 < 0$).

EXAMPLE · P4: LU AND SOLVE

Factor $A = \begin{pmatrix} 1 & 2 \\ 3 & 8 \end{pmatrix}$ as $A = LU$ and solve $Ax = (5, 19)$.

Solution. Pivot 1, multiplier 3: $R_2 \leftarrow (3, 8) - 3(1, 2) = (0, 2)$. So $L = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$, $U = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}$. $Ly = b$: $y_1 = 5$, $y_2 = 19 - 15 = 4$. $Ux = y$: $x_2 = 2$, $x_1 = 5 - 4 = 1$. Check: $(1 + 4,\ 3 + 16) = (5, 19)$. ✓

EXAMPLE · P5: A PIVOT THAT FORCES A SWAP

Factor $A = \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix}$ as $PA = LU$.

Solution. The pivot 0 cannot be used: swap the rows, $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $PA = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$. That is already upper triangular, so $L = I$ and $U = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$.

EXAMPLE · P6: CONDITION NUMBER

$A = \mathrm{Diag}(1, 10^{-3})$, $b = (1, 10^{-3})$, $\delta b = (0, 10^{-3})$. Compare the relative change in x with the bound $\kappa \cdot \|\delta b\|/\|b\|$.

Solution. $x = (1, 1)$, $\tilde{x} = A^{-1}(b+\delta b) = (1, 2)$. Relative change $\|(0, 1)\|/\|(1, 1)\| = 1/\sqrt{2} \approx 0.71$. $\kappa = 1/10^{-3} = 1000$ and $\|\delta b\|/\|b\| \approx 10^{-3}$, so the bound is ≈ 1. A 0.1% change in b changed x by 71%, within the bound.

SECTION 28

Cheat sheet

WEEK 01 ON ONE PAGE

- **Norm**: $\|x\| \geq 0$ ($= 0$ iff $x = 0$), $\|\alpha x\| = |\alpha|\|x\|$, $\|a + b\| \leq \|a\| + \|b\|$. $\ell_1 = \sum |x_i|$, $\ell_2 = \sqrt{\sum x_i^2}$, $\ell_\infty = \max |x_i|$; $\ell_0 =$ number of nonzeros, not a norm.
- $\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n}\|x\|_\infty$, $\|x\|_2 \leq \|x\|_1 \leq \sqrt{n}\|x\|_2$ (sparse $\to$ left equalities, flat $\to$ right).
- **Cauchy–Schwarz**: $|a^T b| \leq \|a\|_2\|b\|_2$, equality iff linearly dependent.
- Tr is cyclic. Eigenvalues: $\det(A - \lambda I) = 0$. Symmetric $\Rightarrow A = U\,\mathrm{Diag}\{\lambda_i\}U^T$, U orthogonal; $\mathrm{Tr} = \sum \lambda_i$, $\det = \prod \lambda_i$.
- $X \succ 0$: $z^T X z > 0\ \forall z \neq 0 \iff$ all $\lambda_i > 0 \iff$ Cholesky $X = LL^T$ exists. 2×2: $a > 0$, $ab - c^2 > 0$. The order $\succeq$ is only partial.
- PSD facts: $A_{ii} \geq 0$, $\det \geq 0$, max entry on the diagonal, $A_{ii} = 0 \Rightarrow$ row i is 0; $\mathrm{Tr}\{XY\} \geq 0$; $X \succeq Y \succ 0 \Rightarrow X^{-1} \preceq Y^{-1}$; $X \succeq Y \Rightarrow MXM^T \succeq MYM^T$.
- $\|A\|_{\mathrm{Fro}} = \sqrt{\sum A_{ij}^2} = \sqrt{\mathrm{Tr}\{AA^T\}}$; $\|A\|_2 = \max_{\|x\|=1}\|Ax\|$ (symmetric: $\max |\lambda_i|$). $\langle A, B\rangle = \mathrm{Tr}\{B^T A\}$; $\mathrm{Tr}\{xy^T\} = x^T y$.
- $\kappa(A) = \|A\|\|A^{-1}\|$; $\dfrac{\|\delta x\|}{\|x\|} \leq \kappa \dfrac{\|\delta b\|}{\|b\|}$.
- **Cholesky**: $A = LL^T$, $\frac{1}{3}n^3$. **LU**: $PA = LU$, $\frac{2}{3}n^3$, partial pivoting. Triangular solve: n^2. Solve: $Ly = Pb$, then $Ux = y$.